

# 06-Data-Model-and-ERD

Data Model & ERD — Leish! v2

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## Data Model & ERD — Leish! v2

| FIELD           | VALUE                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| Document ID     | LEISH-DATA-v2.0                                                                                                               |
| Version         | 2.0.0                                                                                                                         |
| Date            | 2026-08-29                                                                                                                    |
| Source of Truth | <code>src/server/db.ts</code> — <code>PG_SCHEMA</code> (line 207) + <code>SQLITE_SCHEMA</code> (line 477)                     |
| Access Facade   | <code>getDb(): DbFacade</code> — <code>prepare(sql) → Statement {get,all,run}</code> + <code>exec(sql)</code>                 |
| Migration       | <code>scripts/migrate.ts</code> ( <code>npm run db:migrate</code> ) idempotent; SQLite <code>migrateSqlite()</code> auto-runs |

### 1. Overview & Conventions

- **Two-backend facade:** `isPostgres()` → `pg.Pool` (production, `DATABASE_URL`) else `node:sqlite DatabaseSync` ( `LEISH_DB_PATH` or `data/leish.db` , `WAL` + `foreign_keys=ON` ). Placeholders `@name` / `?` translated to `$n` for PG ( `compilePlaceholders` ).
- **IDs:** `TEXT PK` — `randomUUID()` / `crypto.randomUUID()` .

- **Times:** `TEXT` ISO-8601 `new Date().toISOString()` everywhere (no `TIMESTAMPZ` variance across backends).
- **Money:** `INTEGER` sen (cents) — never float. Only `artists.rating` is `REAL`.
- **Booleans:** `INTEGER 0/1` (`verified`, `email_verified`, `consent`, `revoked`, `email_preferences` flags).
- **JSON arrays:** `TEXT` containing `JSON.stringify(...)` (`specialties`, `services`, `portfolio`, `extras`, etc.).
- **Cascade:** `bookings/payments/quotations/messages` `CASCADE` on parent delete (PDPA);  
`admin_audit_log.admin_user_id` `SET NULL` (preserve audit).

drift guard: `detectSchemaDrift()` warns when `PG_SCHEMA` and `SQLITE_SCHEMA` diverge on tables/columns (dev-time only).

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## 2. Entity-Relationship Diagram

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```
admin_audit_log PK(id) FK admin_user_id→users SET NULL
```

Legacy removed scaffold: catalog\_overrides (kept for migration but folded into real columns at seed)

### 3. Table Dictionaries

#### 3.1 USERS

| COLUMN            | TYPE    | CONSTRAINTS                                              | DESCRIPTION                                     |
|-------------------|---------|----------------------------------------------------------|-------------------------------------------------|
| id                | TEXT    | PK, randomUUID()                                         | Public user id ( toPublicUser strips password ) |
| email             | TEXT    | UNIQUE, lowercased                                       | Login key; idx_users_email                      |
| name              | TEXT    | NOT NULL                                                 | Display name (2–80 validated)                   |
| role              | TEXT    | CHECK<br>customer artist studio admin                    | See src/lib/types.ts:1                          |
| password          | TEXT    | NOT NULL                                                 | scrypt hash (+ pepper HMAC if PEPPER_SECRET )   |
| email_verified    | INTEGER | 0/1 default 0, backfilled via migrateSqlite when missing | Verification gate                               |
| consent           | INTEGER | 0/1 default 0                                            | PDPA consent flag                               |
| consent_timestamp | TEXT    | nullable ISO                                             | When consent was given                          |
| created_at        | TEXT    | NOT NULL ISO                                             |                                                 |

**Indexes:** idx\_users\_email(email) . **Backfill:** SQLite adds email\_verified, consent, consent\_timestamp via ALTER TABLE when absent.

### 3.2 ARTISTS (RUNTIME CATALOG)

| COLUMN                                            | TYPE    | CONSTRAINTS             | DESCRIPTION                                                                        |
|---------------------------------------------------|---------|-------------------------|------------------------------------------------------------------------------------|
| <code>id</code>                                   | TEXT    | PK                      | <code>aisha-azman</code> for seeds, UUID for admin-created                         |
| <code>slug</code>                                 | TEXT    | UNIQUE                  | URL key; seeded <code>slug=id</code> ; dedup via <code>slugifyName()+suffix</code> |
| <code>name</code>                                 | TEXT    | NOT NULL                |                                                                                    |
| <code>tagline</code>                              | TEXT    | default ""              |                                                                                    |
| <code>bio</code>                                  | TEXT    | default ""              |                                                                                    |
| <code>image</code>                                | TEXT    | default ""              | URL or <code>/images/...</code>                                                    |
| <code>rating</code>                               | REAL    | default 0               | Blended atomically (2dp)                                                           |
| <code>review_count</code>                         | INTEGER | default 0               |                                                                                    |
| <code>state</code>                                | TEXT    | default ""              | <code>MALAYSIA_STATES</code> value                                                 |
| <code>area</code>                                 | TEXT    | default ""              | within state's <code>AREAS_BY_STATE</code>                                         |
| <code>price_from</code>                           | INTEGER | default 0               | sen; min service price                                                             |
| <code>verified</code>                             | INTEGER | 0/1 default 0           | Admin-only field                                                                   |
| <code>years_experience</code>                     | INTEGER | default 0               |                                                                                    |
| <code>specialties</code>                          | TEXT    | JSON <code>[]</code>    | string[]                                                                           |
| <code>services</code>                             | TEXT    | JSON <code>[]</code>    | <code>Service[]</code><br><code>{name,price,duration}</code>                       |
| <code>bridal</code>                               | TEXT    | JSON <code>[]</code>    | <code>BridalEvent[]</code>                                                         |
| <code>non_bridal</code>                           | TEXT    | JSON <code>[]</code>    | <code>NonBridalEvent[]</code>                                                      |
| <code>availability</code>                         | TEXT    | JSON <code>[]</code>    | string[] slots e.g. "Tomorrow, 10:00 AM"                                           |
| <code>portfolio</code>                            | TEXT    | JSON <code>[]</code>    | image URLs                                                                         |
| <code>referral_code</code>                        | TEXT    | default ""              | Assigned via <code>assignReferralCode</code>                                       |
| <code>referred_by</code>                          | TEXT    | FK artists(id) SET NULL | Referrer                                                                           |
| <code>referral_earnings</code>                    | INTEGER | default 0               | sen                                                                                |
| <code>created_at</code> , <code>updated_at</code> | TEXT    | NOT NULL                |                                                                                    |

Indexes: `idx_artists_state_area(state,area)`.

**3.3 STUDIOS — SAME SHAPE, REPLACES TAGLINE→DESCRIPTION? ACTUALLY BOTH**

**TAGLINE + DESCRIPTION ; SERVICES IS STRING[] NOT SERVICE[] ; PLUS ADDRESS , HOURS , PHONE .**

Full DDL identical to `PG_SCHEMA` block lines 403–424.

**3.4 BOOKINGS**

| COLUMN                           | TYPE    | CONSTRAINTS                                                                              | DESCRIPTION                                                                    |
|----------------------------------|---------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <code>id</code>                  | TEXT    | PK                                                                                       |                                                                                |
| <code>user_id</code>             | TEXT    | FK users CASCADE                                                                         | Customer owner; <code>idx_bookings_user</code>                                 |
| <code>artist_id</code>           | TEXT    | NOT NULL                                                                                 | Catalog artist booked (text, intentionally not strict FK for seed flexibility) |
| <code>studio_id</code>           | TEXT    | FK studios SET NULL, nullable                                                            | Studio variant (Option B); backfilled                                          |
| <code>artist_name</code>         | TEXT    | NOT NULL                                                                                 | Denormalized at booking time for fast display                                  |
| <code>service</code>             | TEXT    | NOT NULL                                                                                 | Service name chosen                                                            |
| <code>price</code>               | INTEGER | NOT NULL                                                                                 | sen; server-derived from <code>artist.services.find(service).price</code>      |
| <code>date</code>                | TEXT    | <code>YYYY-MM-DD</code>                                                                  | Event date ( <code>≥ today</code> at creation)                                 |
| <code>time</code>                | TEXT    | NOT NULL                                                                                 | Slot label ("10:00 AM" etc)                                                    |
| <code>notes</code>               | TEXT    | nullable                                                                                 | ≤2000 chars                                                                    |
| <code>event_type</code>          | TEXT    | nullable                                                                                 | e.g. "solemnization" (backfilled)                                              |
| <code>venue</code>               | TEXT    | nullable                                                                                 | ≤200 chars (backfilled)                                                        |
| <code>guest_count</code>         | INTEGER | default 0                                                                                | 0–1000 (backfilled)                                                            |
| <code>status</code>              | TEXT    | CHECK <code>requested accepted confirmed cancelled completed</code><br>default requested | State machine                                                                  |
| <code>balance_reminder_at</code> | TEXT    | nullable ISO                                                                             | When reminder sent (backfilled)                                                |
| <code>created_at</code>          | TEXT    | NOT NULL                                                                                 |                                                                                |

**Critical constraint:**

```
CREATE UNIQUE INDEX IF NOT EXISTS uq_bookings_slot
ON bookings(artist_id, date, time)
WHERE status IN ('requested', 'accepted', 'confirmed');
```

Prevents double-booking live slots; cancelled/completed don't block.

### 3.5 QUOTATIONS

| COLUMN                                            | TYPE    | CONSTRAINTS                                                        | DESCRIPTION                                      |
|---------------------------------------------------|---------|--------------------------------------------------------------------|--------------------------------------------------|
| <code>id</code>                                   | TEXT    | PK                                                                 |                                                  |
| <code>booking_id</code>                           | TEXT    | FK bookings CASCADE                                                |                                                  |
| <code>base_fee</code>                             | INTEGER | default 0                                                          | sen                                              |
| <code>travel_fee</code>                           | INTEGER | default 0                                                          |                                                  |
| <code>early_call_fee</code>                       | INTEGER | default 0                                                          |                                                  |
| <code>accommodation_fee</code>                    | INTEGER | default 0                                                          |                                                  |
| <code>extras</code>                               | TEXT    | JSON <code>[]</code>                                               | <code>ExtraItem[] {label, amount sen}</code> ≤10 |
| <code>artist_note</code>                          | TEXT    | nullable                                                           | ≤1000                                            |
| <code>total</code>                                | INTEGER | NOT NULL                                                           | <code>quotationTotal()</code> computed sen       |
| <code>status</code>                               | TEXT    | CHECK <code>pending paid expired superseded</code> default pending |                                                  |
| <code>created_at</code> , <code>expires_at</code> | TEXT    | NOT NULL                                                           | <code>expires_at = created_at + 24h</code>       |

### 3.6 PAYMENTS (HYBRID)

| COLUMN                                            | TYPE    | CONSTRAINTS                                                       | DESCRIPTION                                     |
|---------------------------------------------------|---------|-------------------------------------------------------------------|-------------------------------------------------|
| <code>id</code>                                   | TEXT    | PK                                                                | <code>PaymentRecord.id</code>                   |
| <code>booking_id</code>                           | TEXT    | FK bookings CASCADE                                               | <code>idx_payments_booking</code>               |
| <code>type</code>                                 | TEXT    | CHECK <code>deposit balance</code> default deposit                |                                                 |
| <code>amount</code>                               | INTEGER | NOT NULL                                                          | sen                                             |
| <code>currency</code>                             | TEXT    | default MYR                                                       |                                                 |
| <code>provider</code>                             | TEXT    | default dev                                                       | <code>dev billplz</code>                        |
| <code>status</code>                               | TEXT    | CHECK <code>required paid failed refunded</code> default required |                                                 |
| <code>provider_ref</code>                         | TEXT    | nullable                                                          | Billplz <code>id</code> or <code>dev_...</code> |
| <code>provider_url</code>                         | TEXT    | nullable                                                          | Billplz hosted page URL                         |
| <code>created_at</code> , <code>updated_at</code> | TEXT    | NOT NULL                                                          |                                                 |



**Unique:** `CREATE UNIQUE INDEX uq_payments_booking_type ON payments(booking_id,type)` — one deposit + one balance per booking. **Migration:** legacy single-`UNIQUE(booking_id)` tables rebuilt to this shape; backfill adds `type` and `provider_url` if missing.

### 3.7 `PAYOUTS`

| COLUMN                      | TYPE    | CONSTRAINTS                                                  | DESCRIPTION                                       |
|-----------------------------|---------|--------------------------------------------------------------|---------------------------------------------------|
| <code>id</code>             | TEXT    | PK                                                           |                                                   |
| <code>artist_user_id</code> | TEXT    | FK users SET NULL                                            | Claimed artist user (nullable when unclaimed)     |
| <code>booking_id</code>     | TEXT    | FK bookings CASCADE                                          | one payout per booking                            |
| <code>gross_sen</code>      | INTEGER | NOT NULL                                                     | quotation.total                                   |
| <code>commission_sen</code> | INTEGER | NOT NULL                                                     | computed via <code>computeCommission</code>       |
| <code>net_sen</code>        | INTEGER | NOT NULL                                                     | <code>max(0, gross - commission - deposit)</code> |
| <code>status</code>         | TEXT    | CHECK <code>pending settled failed</code><br>default pending |                                                   |
| <code>settleable_at</code>  | TEXT    | nullable ISO                                                 | <code>eventDate+24h</code>                        |
| <code>settled_at</code>     | TEXT    | nullable ISO                                                 | set when <code>settled</code>                     |
| <code>notes</code>          | TEXT    | nullable                                                     | "Commission waived (small booking)" etc           |
| <code>created_at</code>     | TEXT    | NOT NULL                                                     |                                                   |

**Indexes:** `idx_payouts_status`, `idx_payouts_artist`. **Cardinality:** ≤1 per `booking_id` (app guard; could be made `UNIQUE` in future).

### 3.8 `SESSIONS` (JTI REVOCATION)

| COLUMN                  | TYPE    | CONSTRAINTS      | DESCRIPTION                           |
|-------------------------|---------|------------------|---------------------------------------|
| <code>jti</code>        | TEXT    | PK               | JWT ID                                |
| <code>user_id</code>    | TEXT    | FK users CASCADE | owner; <code>idx_sessions_user</code> |
| <code>revoked</code>    | INTEGER | 0/1 default 0    | revoked=1 means blocked               |
| <code>expires_at</code> | TEXT    | NOT NULL         | <code>exp</code> mirrored for cleanup |
| <code>created_at</code> | TEXT    | NOT NULL         |                                       |

### 3.9 REVIEWS

| COLUMN                   | TYPE    | CONSTRAINTS                        | DESCRIPTION                                        |
|--------------------------|---------|------------------------------------|----------------------------------------------------|
| <code>id</code>          | TEXT    | PK                                 |                                                    |
| <code>entity_type</code> | TEXT    | CHECK <code>(artist studio)</code> | polymorphic target                                 |
| <code>entity_id</code>   | TEXT    | NOT NULL                           | <code>artists.id</code> or <code>studios.id</code> |
| <code>booking_id</code>  | TEXT    | UNIQUE FK bookings CASCADE         | gates: one review per booking                      |
| <code>user_id</code>     | TEXT    | FK users SET NULL                  | author; nullable for seed rows                     |
| <code>author_name</code> | TEXT    | NOT NULL                           | denormalized display name                          |
| <code>rating</code>      | INTEGER | CHECK 1-5                          | rounded int                                        |
| <code>event</code>       | TEXT    | nullable                           | service/event label                                |
| <code>text</code>        | TEXT    | NOT NULL                           | review body                                        |
| <code>created_at</code>  | TEXT    | NOT NULL                           |                                                    |

Index: `idx_reviews_entity(entity_type,entity_id)` .

### 3.10 REFERRALS

| COLUMN                                           | TYPE    | CONSTRAINTS                                                 | DESCRIPTION      |
|--------------------------------------------------|---------|-------------------------------------------------------------|------------------|
| <code>id</code>                                  | TEXT    | PK                                                          |                  |
| <code>referrer_type</code>                       | TEXT    | CHECK <code>(artist studio)</code>                          | who referred     |
| <code>referrer_id</code>                         | TEXT    | NOT NULL                                                    |                  |
| <code>referee_type</code>                        | TEXT    | CHECK <code>(artist studio)</code>                          | who was referred |
| <code>referee_id</code>                          | TEXT    | NOT NULL                                                    |                  |
| <code>status</code>                              | TEXT    | CHECK <code>(pending qualified paid)</code> default pending |                  |
| <code>reward_sen</code>                          | INTEGER | default 0                                                   | earned           |
| <code>qualified_at</code> , <code>paid_at</code> | TEXT    | nullable ISO                                                |                  |
| <code>created_at</code>                          | TEXT    | NOT NULL                                                    |                  |

Indexes: `idx_referrals_referrer` , `idx_referrals_referee` , `idx_referrals_status` .

### 3.11 MESSAGES

| COLUMN                  | TYPE | CONSTRAINTS         | DESCRIPTION                       |
|-------------------------|------|---------------------|-----------------------------------|
| <code>id</code>         | TEXT | PK                  |                                   |
| <code>booking_id</code> | TEXT | FK bookings CASCADE | <code>idx_messages_booking</code> |
| <code>sender_id</code>  | TEXT | FK users CASCADE    |                                   |
| <code>body</code>       | TEXT | NOT NULL            | message text                      |
| <code>created_at</code> | TEXT | NOT NULL            | ordered ASC for replay            |

### 3.12 COMMUNICATION & EMAIL TABLES

| TABLE                            | KEY COLUMNS                                                                                                                                                    | PURPOSE                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <code>email_outbox</code>        | <code>id, to_email, subject, text, html, created_at</code>                                                                                                     | de<br>pr<br>sin<br>vie<br>at<br>em |
| <code>email_retries</code>       | <code>id, to_email, subject, text, html, attempts, max_attempts=3, next_retry, last_error, created_at</code>                                                   | As<br>re<br>qu<br>to<br>at         |
| <code>email_preferences</code>   | <code>user_id PK FK, booking_created, quotation_sent, invoice_sent, quotation_expiry, balance_reminder, status_changed INTEGER 0/1 default1, updated_at</code> | Pe<br>to                           |
| <code>email_verifications</code> | <code>id, user_id FK CASCADE, token_hash, expires_at, used_at, created_at</code>                                                                               | Si<br>us<br>ha<br>to<br>ve         |
| <code>password_resets</code>     | <code>id, user_id FK CASCADE, token_hash, expires_at, used_at, created_at</code>                                                                               | Si<br>1h<br>re                     |

### 3.13 OPERATIONAL TABLES

| TABLE                          | KEY COLUMNS                                                                                                                                                                                                                                            | PURPOSE                                                                                                                            |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>artist_profiles</code>   | <code>user_id</code> PK FK CASCADE, <code>artist_id</code> TEXT<br><code>UNIQUE</code> , <code>claimed_at</code>                                                                                                                                       | Artist/studio claims — scope filter                                                                                                |
| <code>studio_profiles</code>   | <code>user_id</code> PK FK CASCADE, <code>studio_id</code> TEXT<br><code>UNIQUE</code> , <code>claimed_at</code>                                                                                                                                       | Same for studios                                                                                                                   |
| <code>platform_settings</code> | <code>key</code> PK, <code>value</code> TEXT, <code>updated_by</code> FK SET NULL,<br><code>updated_at</code>                                                                                                                                          | Runtime knobs:<br><code>booking_fee_sen</code> ,<br><code>commission_rate_bps</code> ,<br><code>commission_waiver_sen</code>       |
| <code>admin_audit_log</code>   | <code>id</code> PK, <code>admin_user_id</code> FK SET NULL, <code>action</code> ,<br><code>target_table</code> , <code>target_id</code> , <code>details</code> JSON '{}',<br><code>created_at</code>                                                   | Audit trail; indexes on<br><code>admin_user_id</code> , <code>created_at</code>                                                    |
| <code>catalog_overrides</code> | <code>id</code> PK, <code>entity_type</code> , <code>entity_id</code> , <code>field</code> ,<br><code>value</code> , <code>updated_by</code> , <code>created_at</code> , <code>updated_at</code> ,<br><code>UNIQUE(entity_type,entity_id,field)</code> | Legacy — folded into real columns<br>at seed; kept for migration but<br><code>idx_catalog_overrides_entity</code><br>still indexed |

**Note** on FKs in tests: `admin_audit_log.admin_user_id → users(id)` has real FK; tests must seed a user row before inserting audit.

## 4. Constraints & Indexes (Summary)

| CONSTRAINT / INDEX                        | SQL EXCERPT                                                    | PURPOSE                                     |
|-------------------------------------------|----------------------------------------------------------------|---------------------------------------------|
| <code>uq_bookings_slot</code>             | <code>UNIQUE(artist_id,date,time) WHERE status IN (...)</code> | Anti double-booking                         |
| <code>uq_payments_booking_type</code>     | <code>UNIQUE(booking_id,type)</code>                           | One deposit + one balance                   |
| <code>uq_artist_profiles_artist_id</code> | <code>UNIQUE(artist_id)</code>                                 | One owner per artist                        |
| <code>uq_studio_profiles_studio_id</code> | <code>UNIQUE(studio_id)</code>                                 | One owner per studio                        |
| <code>reviews.booking_id UNIQUE</code>    | Column constraint                                              | One review per booking                      |
| Role checks                               | <code>CHECK role IN (...)</code> etc                           | Enforce vocabularies                        |
| State checks                              | <code>CHECK status IN (...)</code>                             | Enforce booking/quote/payment/payout states |
| Range checks                              | <code>CHECK rating BETWEEN 1 AND 5</code>                      | Validate rating                             |
| Foreign keys                              | <code>REFERENCES ... ON DELETE CASCADE/SET NULL</code>         | PDPA & audit semantics                      |
| Area index                                | <code>idx_artists_state_area(state,area)</code>                | Fast catalog pre-filter                     |
| Booking user index                        | <code>idx_bookings_user(user_id)</code>                        | Customer booking list                       |
| Payment booking index                     | <code>idx_payments_booking(booking_id)</code>                  | Serialization join                          |

## 5. Money Logic & Derived Fields

**Quoting** ( `src/server/quotations.ts:44` ):

```
function quotationTotal(input): number {
  return input.baseFee + (travelFee??0) + (earlyCallFee??0) + (accommodationFee??0) + sum(extras.amou
nt)
```

Stored in `quotations.total` ; never float.

**Commission** ( `src/server/settings.ts:100` ):

```
function computeCommission(totalSen, rateBps, waiverSen) {
  const waived = total < waiverSen;
  const commissionSen = waived ? 0 : Math.round(total * rateBps / 10000);
  return { totalSen, waived, commissionSen, artistNetSen: total - commissionSen };
}
```

**Payout net** ( `src/server/payouts.ts:79` ): `artistReceivesSen = max(0, artistNetSen - depositSen)` where `depositSen = getBookingFeeSen()`.

**Booking serialization** ( `src/app/api/bookings/route.ts:39` ): `balanceDueDate = eventDate - 3 days`, `balanceAmount = max(0, total - fee)` when quotation exists and not expired.

## 6. Seeding & Migration

### 6.1 CATALOG SEEDING (IDEMPOTENT)

- Source: `SEED_ARTISTS` (7 entries) + `SEED_STUDIOS` (4) `src/lib/data.ts`.
- Runner: `ensureCatalogSeeded()` called lazily before first catalog read AND explicitly via `npm run db:seed-catalog` ( `scripts/seed-catalog.ts` ).
- Logic: Inserts new `artists/studios` rows when missing; legacy `catalog_overrides` rows are **folded** into real columns and deleted (so old overrides don't drift).
- Placeholders: Admin-created artists without image get `/images/hero.jpg` to satisfy `next/image` requirements.

### 6.2 ADMIN SEEDING

- `scripts/seed-admin.ts` : reads `ADMIN_EMAIL/ADMIN_PASSWORD`, upserts `users` ( `role='admin'` ), hash via `scrypt`; idempotent upgrade of existing non-admin user.

### 6.3 SCHEMA MIGRATIONS

- **Postgres**: `PG_SCHEMA` applied lazily on `getDb()` first query via `pool.query(PG_SCHEMA)` in `pgReady`; explicitly via `scripts/migrate.ts`.
- **SQLite**: `SQLITE_SCHEMA` + `migrateSqlite(db)` additive `ALTER TABLE` + partial-index creation; supports aging DBs lacking newer columns ( `email_verified`, `provider_url`, `type`, etc.).
- Both backends apply `CREATE UNIQUE INDEX IF NOT EXISTS uq_payments_booking_type` separately to avoid rollback of entire schema on legacy databases.

## 7. Data Retention & Lifecycle

Per `docs/PDPA_RETENTION_GUIDELINES.md`:

| ENTITY                                         | RETENTION                                                                                             | DELETION                                                                                        |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>users</code>                             | Until account deletion; then anonymized<br>audit log retained with<br><code>admin_user_id→null</code> | <code>DELETE /api/me</code> cascades bookings/<br>messages/payments via FK <code>CASCADE</code> |
| <code>bookings/quotations/<br/>payments</code> | 7 years for financial audit (configurable)                                                            | Cascade on user delete; admin may<br>override status but never hard-delete<br>for audit         |
| <code>messages</code>                          | Same as booking                                                                                       | Cascade                                                                                         |
| <code>reviews</code>                           | Indefinite (anonymized if user deleted →<br><code>user_id null</code> kept)                           | No hard delete of public proof                                                                  |
| <code>sessions</code>                          | Expire after <code>expires_at</code> ; gc via cron or<br>lazy cleanup                                 | <code>revoked</code> rows inert                                                                 |
| <code>email_outbox</code>                      | Dev only; not retention-bound                                                                         | Ephemeral                                                                                       |

## 8. Verification of Model

| CHECK                     | METHOD                                                                                                                                                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PG_SCHEMA vs SQLite drift | <code>extractSchemaTables()</code> + <code>detectSchemaDrift()</code> warning<br>(dev) — run <code>npm run db:migrate</code> and compare<br><code>PG_SCHEMA</code> / <code>SQLITE_SCHEMA</code> tables+columns |
| Placeholder translation   | Unit tests <code>compilePlaceholders/resolveParams</code>                                                                                                                                                      |
| FK enforcement            | Integration tests with <code>:memory:</code> SQLite <code>foreign_keys=ON</code>                                                                                                                               |
| Money integer             | Lint grep for float amounts in <code>src/server/<br/>{payments,quotations,payouts}.ts</code>                                                                                                                   |
| Index coverage            | <code>EXPLAIN QUERY PLAN</code> on <code>listArtists(state,area)</code> must<br>use <code>idx_artists_state_area</code>                                                                                        |

Next: `docs/07-API-Specification.md` details HTTP contracts over this model; `docs/08-Traceability-Matrix.md` maps requirements to tables and endpoints.